

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1
PAPER 1 November 2020
2 hours 30 minutes

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Evaluate $5 + 5 \times 4 - 5$. [2]
2. Evaluate $-6 - (-15)$. [3]
3. Work out $1/5 + 1/4$. Give your answer as a mixed number if possible. [3]
4. Find 10% of 80. [3]
5. Solve $3x + 7 = 28$. [3]
6. Share \$44 in the ratio 4 : 7. [4]
7. Find (a) the HCF and (b) the LCM of 18 and 42. [4]
8. Write 800000 in standard form. [3]
9. Simplify $5^2 \times 5^4$. Leave your answer as a power of 5. [3]
10. $n(A) = 16$, $n(B) = 15$ and $n(A \cap B) = 3$. Find $n(A \cup B)$. [3]
11. Expand and simplify $5(x + 1) - 3x$. [3]
12. Factorise completely $12x + 48$. [3]
13. Factorise $x^2 + 8x + 12$. [4]
14. Make h the subject of the formula $A = 2\pi r(r + h)$. Take π as a symbol. [4]
15. Solve $3x + 1 < 16$. [3]
16. Solve the simultaneous equations
$$\begin{aligned}x + y &= 9 \\x - y &= -1.\end{aligned}$$
[4]
17. The n th term of a sequence is $3 + 3(n - 1)$. Find the 8th term. [3]
18. In triangle ABC, angle A = 54° and angle B = 43° . Find angle C. [3]
19. A regular polygon has 12 sides. Find the size of each exterior angle. [3]
20. A right-angled triangle has shorter sides 24 cm and 45 cm. Calculate the hypotenuse. [4]
21. In right-angled triangle PQR, angle P = 30° and hypotenuse PR = 8 cm. Calculate PQ, the side opposite 30° . [4]

- 22.** A triangle has base 14 cm and perpendicular height 5 cm. Find its area. **[3]**
- 23.** Calculate the area of a circle of radius 14 cm. Take $\pi = 22/7$. **[4]**
- 24.** A cylinder has radius 7 cm and height 10 cm. Find its volume. Take $\pi = 22/7$. **[4]**
- 25.** Find the gradient of the line joining (5, 5) and (7, 7). **[3]**
- 26.** Find the midpoint of the line joining (5, 5) and (7, 7). **[3]**
- 27.** A bag contains 3 red, 3 blue and 5 green beads. A bead is drawn at random. Find the probability that it is red. **[3]**
- 28.** Find the mean of 6, 4, 6, 8, 12. **[3]**
- 29.** Work out $\begin{pmatrix} 2 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \end{pmatrix}$ where $\begin{pmatrix} \\ \end{pmatrix}$ denotes a 2×2 matrix times a column vector. **[4]**
- 30.** Given $u = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $v = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, find $3u + v$. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).